DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY LUCKNOW

STUDY & EVALUATION SCHEME& SYLLABUS

FOR

B. TECH 4th YEAR
Carpet & Textile Technology

ON

CHOICE BASED CREDIT SYSTEM

(EFFECTIVE FROM THE SESSION: 2019-20)

SEVENTH SEMESTER
CARPET AND TEXTILE TECHNOLOGY

Sl No.	Subject Code	Subject Name	Teaching Deptt.	L-T-P	Th/Lab Marks	Sessional		Total	Credit
					ESE	CT	TA		
1		**OPEN ELECTIVE COURSE I**	Other Deptt.	3—0---0	70	20	10	100	3
2		**DEPTT ELECTIVE III**	Core Deptt.	3—0---0	70	20	10	100	3
	RCT070	MILL PLANNING AND ORGANISATION							
	RCT071	ENGINEERING OF TEXTILE STRUCTURE							
3		**DEPTT ELECTIVE IV**	Core Deptt.	3—1---0	70	20	10	100	4
	RCT075	CARPET ANALYSIS & TESTING							
	RCT076	HOME TEXTILE MANUFACTURE II							
	RCT077	CARPET DESIGN							
4	RCT701	ADVANCE YARN MANUFACTURE	Core Deptt.	3—1---0	70	20	10	100	4
5	RCT702	STRUCTURAL PROPERTIES OF FIBRES	Core Deptt.	3—0---0	70	20	10	100	3
6	RCT751	ADVANCE YARN MANUFACTURE LAB	Core Deptt.	0—0---2	50		50	100	1
7	RCT752	CARPET AND HOME TEXTILES II LAB	Core Deptt.	0—0---2	50		50	100	1
8	RCT753	INDUSTRIAL TRAINNING	Core Deptt.	0 - 0 - 3			100	100	2
9	RCT754	PROJECT-1	Core Deptt.	0—0---6			200	200	3
	TOTAL				450	100	450	**1000**	**24**

Specialization : ADVANCE CARPET TECHNOLOGY,
HOME TEXTILE TECHNOLOGY,
TEXTILE DESIGN TECHNOLOGY

MILL PLANNING & ORGANISATION 3-0-0 CREDIT 3

UNIT 1

Selection of site for Textile/Carpet mills, General location, Calculation of spatial requirements, lighting of industrial building, factors influencing site selection, Climatic considerations, Humidification its importance and use in textile/carpet industry.

Unit2

Balancing a layout of cotton spinning/woolen spinning department ,Prepration of project report ,Breakeven point and evaluation of project work. Calculation for no. of machines in spinning /spin plan - Preparation of organizationfor ring spinning mill and preparatory, departments based on ring spindle capacityand production of ring spun yarn.

Assumptions, Machinery Organizations, Requirement of Miscellaneous FixedAssets& Machinery Stores & Spares, Requirement& Calculations related toElectrical Power, Lighting etc.
Unit 3

Balancing of machines and layout of weaving department. Type of industry: JV , company limited, public limited companies.

Unit 4

Balancing of machine for wet processing of textiles industry ,layout of processing department ,Water, Steam, Compressed Air

Techno-economic Viability - Calculations of cost of project – Means of Finance –

Estimates of sales & production – cost of production – working capital requirement – Profitability Projection – Break even point – Projected cash flow statements.

Unit 5

Site selection of carpet industry. Balancing of machine and layout of carpet industry , piece & time wage system ,incentives ,export documentation of carpets.

References:

1. Textile Project Management by A. Ormerod, The Textile Institute Publication.
2. Project, Planning Analysis, Selection Implementation & Review by Prasanna Chandra, Tata McGraw Hill Publishing Co. Ltd.,
3. Management of Textile Production, A. Ormorod. Newnes – Butter WortrsPublication.
4. Plant location, Layout & Maintenance by Ruddele Reed.

Industrial Organisation &Engg. Economics T.R. Banga& S.C. Sharma, KhannaPublishers, Delhi.

ENGINEERING OF TEXTILE STRUCTURE 3-0-0 CREDIT 3

UNIT–I [8]

Yarn geometry-idealised yarn geometry, relationship of yarn number and twist factor. Twist contraction, limit of twist.

Packing of fiber in yarn . Ideal packing, hexagonal close packing and to other forms. Packing factor and its measurement. Yarn diameter.

UNIT–II [8]

Fiber migration- mechanism of migration, condition for migration to occur, frquency of migration, migration in blended yarns.

Mechanics of staple fibre yarns, the practical and experimental studies. Mechanics of staple fibre yarns, Hambureger model and later modifications. Spinnability and torsional behavior of Fibres and yarns.

UNIT–III [8]

Translation of fibres properties into yarn properties, extension of continous filament yarn for small strains and large strains; prediction of breakage.

UNIT – IV [8]

Elements of fabric geometry, cloth setting theories. Fabric cover and fractional cover, fabric cover and fabric weight relationship. Fabric firmness,

UNIT – V [8]

Pierce concept of fabric geometry and its application, Flexible and elastic threads model, crimp interchange & crimp balance equation

Uniaxial and biaxial tensile behavior of cloth.

Reference

1. Textile yarn by Goswami
2. Textile mathematics by J.E. Booth
3. Pierce papers in Journal of textile institute 1930, 1937
4. Watson’s textile design
5. AATCC review, October , 8 (10), 30-34

CARPET ANALYSIS & TESTING 3-1-0 CREDIT 4

UNIT I [8]

Need for carpet testing. Different aspects of quality testing & performance assessment of carpets, Norms for various performance parameters of carpet

UNIT – II [8]

Testing of functional properties of carpet and floor coverings- (a) Appearance retention (b) Carpet durability including Soilability, carpet abrasion resistance (c) Resilience (d) Tendency of pilling and fuzzing (e) Other properties like insulation properties, acoustic properties, electrostatic properties etc.

UNIT – III [8]

Brief Description and principle operation of following Carpet Testing Equipments:
Dynamic Loading Machine, Tuft Withdrawal Tensometer, Pilfuz Carpet Tester, Usometer, Hexapod Tumble Tester, Courtaulds Tetrapod Walker

UNIT – IV [8]

Brief Description and principle operation of following Carpet Testing Equipments:
Digital Thickness Gauge, Portable Carpet Thickness Gauge, Drum Testing Device, Roller Chair Testing Device, Carpet Static Loading Device, Carpet Wear and Abrasion Tester, Types of Carpet Flammability Tester

UNIT-V [8]

Carpet analysis and reproduction for knotted, tufted pile carpets, flat woven daries, table tufted , shaggy. Cross section analysis of loom made pile carpet- v tuft , w-tuft.

Grand total lectures required=42

References:

1. IWS Test Standard 1, 2, 3
2. Manuals of Carpet Testing of SDL & WIRA, U. K.
3. Carpet Manufacture by Crawshaw

HOME TEXTILE MANUFACTURE II 3-1-0 CREDIT 4

UNIT I [8]

Fullness: Definition, Darts, tucks, pleats. Flares, Godets, Gathers, Shirrs and Frills or Ruffles.
Placket finishes: Definition, classification, continuous bound, bound and faced (two piece) plackets, zipper placket. Tailored placket.
Embroidery: An Introduction, Various materials used in embroidery, Different types of Embroidery Stitches, Precaution during setting the fabric on frame.

UNIT-II [8]

Components & Trims: Labels and Motifs , Linings Interlining, Wadding ,Lace, Braids & Elastics , Seam Binding & Tape , Shoulder Pad , Fasteners, performance properties of Components and trims.
Darning: Definition and various tools used in darning.
Patch Work: Definition, Different styles of Patchwork Techniques such as Pieced Patchwork, Shell Patchwork, Suffolk Puffs, Crazy Patchwork, Log Cabin Patchwork, Strip Patchwork, Seminole Patchwork, Folded Star Patchwork, Mayflower Patchwork and Pleated Patchwork.

UNIT-III [8]

Appliqué: Definition, Various Styles of Appliqué Techniques, Standard appliqué, Appliqué Perse, Reverse appliqué, Padded appliqué, folded appliqué, Shadow appliqué, Lace appliqué
Quilting: Definition, Various Styles of Quilting, Wadded Quilting, Padded Quilting, Corded Quilting, Shadow Quilting.

UNIT-IV [8]

Curtains: - Definition, Various Styles, Choices of Fabrics, Calculating the amount of material needed, Manufacturing Steps of Curtains, Casings, Methods of furnishing draperies at the top with pleats .Use of drapery rods, Hooks, tapes rings and pins.
Pillows: Types of Pillows, Knife-edge Pillow, Box-edge Pillow, Basic Measurements, Constructing a Knife-edge Pillow Covers, Tufting Pillows, Pillow Shams, Ruffled Pillow, Pillow Sham with flat self-border, Box-edge Pillow Cover: Rectangular box-edge cover, Boxed effect without boxing strip, Circular box-edge cover, Bolster and its covers: Round bolster and Wedge bolster.

UNIT-V [8]

Bedspreads: Estimating yardage for a spread, bedspread types and its manufacturing: throw, flounced and tailored.
Blankets: Definition, blankets types.
Terry Towel: - Definition, The Parts of a Conventional Terry Towel, Classification of Terry Towels, Structure of a Towel, Physical Properties of a Towel, Quality Defects, Technology of terry towel production.Kitchen textiles – features and applications

Grand total lectures required=40

Reference:-

1. Soft furnishing book by Kartin Cargill, Reed consumer books limited, London.
2. Soft furnishing by saarah Campbell and Hilary More, MacDonald books, QED publishers Limited London.
3. Simplicity's (1993) –Simply the best home decoration book, A fire side book as published by Simon and Schulster (New York). London. The simplicity pattern company Inc.
4. Home Fashion
5. Cloths line (Journal)
6. House & Garden (Journal)
7. Textiles Para El Hogar (Journal) Distribution &Suibscription – Ecuadar, 75, entresuelo, 08029 Barcelona, Espane, e.mail: publica@publica.es, castellon@publica.es
8.process control in home textiles manufacturing K KGoswami, Abhishek publishing ,.Chandigarh. (In press)
9. Fundamentals of dress by Kettunen,Marietta, Mc.Graw Hill Book Company,New York.
10. Practical Dress Design -Principles of Fitting and pattern making by Mabel.D.Erwin, Macmillan Co., New York.
11. Basic Process of Clothing Construction by Doongali D.S. Deshpande,R., NewRaj Book Depot, New Delhi.
12. Fashion Encyclopedia by Catherine Houck., St.martin's Press, Newyork.
13. Betty Foster's - Fashion maker by Betty Foster, Heinemann Professionalpublishing Limited,- Oxford.

CARPET DESIGN 3—1—0 CREDIT 4

Unit-1 [5]

Introduction of Computer Aided Designing, Creation of Design direct on Computer Screen by using CAD tools (Mouse / Digitizer), Arrangement & layout of Motif to form Design, vector and raster images, x and y in designs.

Unit-2 [5]

Customizable settings, views, new designing creations, Scanned photographs editing, File Utilities, Freehand tools,Geometric tool group, selection group, zoom group, selection utilities, General group.

Unit-3 [10]

Creating motifs in computer: drawing tools, motif scanning, scanning parameters, editing the image for graph making –scaling, rotating, reversing, convert to full methods of different style and forms of design using computer i.e. resize, group, irregular scale, normal scale, covert to full, drop repeat, exchange horizontally, vertical exchange adjust repeat.

Unit-4 [10]

Importance of color application in motif, Colour utilities: - colour protection, colour separation, transparent colour, Changecolours, colour reduction, colour reduction based on similarity. Tracing a graph/Design Plate Print out, woolconsumption print out.Colour Library – overview, creating new colour library, DPI calculations.

Unit-5 [10]

Preparation of computerized graph design from edited motif with suitable weaves, Weavecreation, creation of weaves and saving, Various steps in transferring designs from sketch to Point paper, Pixel resolution and its relation with threads and thread per inch, Creation of Different texture with the help of CAD, Comparison of Manual graph Making and Computer Aided graph Making.

Grand total lectures required=42

Reference:

1. Manual for Autotex software (PLC consultancy)
2. User Manual for Texcells (Ned graphics)
3. Computer Aided Textile Design, II of HT, Salem (Manual
4. Annual World, Computers in the world of textiles, Textile Institute, UK, 1984.Taylor P,
5. The Textile Institute. Winning through Information Technology, UK.
6. Berkstresser. Buhanan&Graddy, Automation in the Textile Industry: from Fibres to Apparels, The Textileinstitute, UK.1995
7. Veinsinet D O. Computer Aided Drafting & Design-Concept & Application, 1987
8. PhirozDastoor, " Application of CAD in the Industrial Fabrics ",Journal of the Textile Institute Part -111,Manchester, 1993.
9. Aldrich, W. (Ed), " CAD in clothing and textiles: A collection of experts view ",Blackwell Science, 2nd Edition, U.K., 1994.

Jayaraman, S., " Computer Science and Textile Science ", T.P.Vol.26, No.3, Textile Inst., Manchester, 1995.

Advance Yarn Manufacture L:T:P::3:1:0 CREDIT 4

UNIT-I **[8]**

Rotor Spinning: Principle of Operation; brief description of various components, formation of yarn; yarn withdrawal & winding;
Limitation of rotor spinning process, Numerical calculation in Rotor spinning process

UNIT – II **[8]**

Friction Spinning: Principle of operation, brief description & process calculation of DREF 2& DREF – 3.

UNIT III **[8]**

Air Jet Spinning:Principle of formation of yarn in MurataAir jet spinning, Properties of Air jet spun yarn & the factor influencing air jet spun yarn.

UNIT-IV **[8]**

Brief discussion on Plyfil spinning system. Study of Principle ofoperation
- Self Twist Spinning
- Twist less Spinning.
- Wrap Spinning

Properties of yarn and limitation of these spinning systems.

UNIT V **[8]**

Textured yarn technology, various methods used for Texturing like Airjet and False twist process.

Grand total lectures required=42

References-

1. Eric Oxtoby, Spun Yarn Technology, Butterworths, London
2. R. V. MahendraGowda, New Spinning Systems, NCUTE Publications
3. W. Klein, New Spinning systems
4. Textured Yarn Technology, by J.W.S. Hearle

STRUCTURAL PROPERTIES OF FIBRES (L-T-P 3-0-0) Credits: 3

UNIT I **[8]**

Identification of fibres, solvents of fibres. Microscopic view of fibres. Blend analysis of mixture fibres as per standard methods.

UNIT II **[8]**

Differential heat of sorption, integral heat of sorption, Moisture absorption, effect of hydrophilic groups, moisture absorption in crystalline and non crystalline region, directly and indirectly attached water.

UNIT III **[8]**

Mechanical properties of fibres, Relation between structure and mechanical properties of fibres, Basic mechanical properties (tenacity elongation, modulus, work of rupture, Elastic recovery, time effects.

UNIT IV **[8]**

Thermal behaviour of textile fibres by DSC, TGA, Thermal Mechanical Analysis, Density Gradient Column, Preparation of density gradient column.

UNIT V **[8]**

Optical properties of fibres, Birefringence behaviour, dielectric properties, fibre friction, fibre friction measurement and static charge measurement.

Reference Book:

1. Manufactured fibre technology by V.B. Gupta, V.K. Kothari
2. Physical properties of fibre by J.W.S. Hearle
3. Thermal behavior of material by Turi
4. Modern yarn production by Ray
5. Textile fibres by ATIRA
6. ASTM Standard books

Advance Yarn Manufacture Lab L:T:P::0:0:2 CREDIT 1

1. To study different parts & their functions of a TFO m/c.
2. Draw the gearing diagram of VJ-150HS TFO & determine the TPI of the yarn & twist constant of the machine.
3. Draw the gearing diagram of VJ-150HS TFO & determine the take up roller speeds in terms of m/min & productivity of the m/c.
4. Draw the gearing diagram of the transmission of drive of the take up roller of VJ-150HS TFO & determine the overfeed % & the m/c constant for overfeed %.
5. Draw the gearing diagram of Ring doubling m/c & determine grams per spindle for a given count.
6. To study different parts a DREF 2000 m/c & study their functions.
7. To calculate Production per head of DREF2000 m/c.
8. To study different parts a Rotor m/c & study their functions. To calculate Production per head of Rotor m/c.

CARPET AND HOME TEXTILES II LAB 0-0-2 CREDIT 1

A. Determination of tuft withdrawal force using tuft withdrawaltensometer.
Determination of thickness of carpet and carpet backing using portable thicknessgauge.

Determination of dimensional stability ofcarpet

Measurement of pile height of carpet using leafgauge.

Determination of compressibility and % recovery of carpet by Digital thicknessgauge.

Determination of Abrasion resistance and weight loss incarpet.

Determination of appearance retention of acarpet.

Determination of Rubbing fastness of carpet as per ISmethod

Determination of Water fastness of carpet as per ISmethod

Determination of total pile weight of carpet as per standardmethod

B. Preparation of samples for following fullness tools:-
Darts.Tucks.Pleats.Godets.Gathers.Shirrs.Frills.

Preparation of samples for different types ofPlackets.

Sample preparation for hand embroiderystitches.

(Running, Stem, Single lace, Double lace, Chain, Satin, Lock, Whip, Cross, Pipe, Loop, Flat, Knotted).
Preparation of samples for different types ofpatchworks.

Preparation of samples for different types ofappliqués.

Preparation of samples for different types ofquilting.

Prototypes development of various home textileproducts.

Embroidering any one furnishing item (table cloth / Table runners /Pillowcovers

/Cushion cover / others)

C. Drawing and painting Equipments&Tools.
Geometrical structures, Exploration of forms, Shapes & line with in the natural forms or objects. Still Lifedrawings

Stripes & Checks effect onfabric,

Colour&DesignCreations: Dhurries- 30^{ct},60^{ct}, 80^{ct} And Boxes & Round Compositions, Colour Wheel, Concept of shadetone,

Carpetdesigns:

Tufted - Floral& ModernDesigns,
Tibbetan – Modern geometrical & Floral, Converting Natural form of designs into Abstract, Modern & Contemporary,
PERSIAN/Traditional Designs,
Colour ForecastingIdeas.

NOTE:
Experiments shall be decided on factors like:

- Facilities installed atInstitute.
- Accessibility to Industry & nearbyInstitutes.
- Trend of Technological Developments in National & Internationalperspective.

RCT753 Industrial Training (0—0—3) CREDIT 2

Three month training in carpet /textile industry. Details of man, machine, material involved in the industry. Production layout , process layout organisation chart etc. The report must be in hard bound form and a ppt presentation to be made for evaluation**.**

RCT754 Project-1 (0—0—6) CREDIT 3

EIGHTTH SEMESTER
CARPET AND TEXTILE TECHNOLOGY

Sl No.	Subject Code	Subject Name	Teaching Deptt.	L-T-P	Th/Lab Marks	Sessional		Total	Credit
					ESE	CT	TA		
1		**OPEN ELECTIVE II**	Other Deptt.	3—0---0	70	20	10	100	3
2	RCT080 RCT081	**DEPTT. ELECTIVE V** TEXTURED YARN TECHNOLOGY (NPTEL/MOOCS) ADVANCE FABRIC STRUCTURE	Core Deptt.	3—1---0	70	20	10	100	4
3	RCT085 RCT086 RCT087	**DEPTT. ELECTIVE VI*** ADVANCE TEXTILE PRINTING TECHNOLOGY (NPTEL/MOOCS) QUALITY CONTROL IN HOME TEXTILES DESIGN MANAGEMENT	Core Deptt.	3—0---0	70	20	10	100	3
4	RCT851	GD & SEMINAR	Core Deptt.	0—0---3			100	100	2
5	RCT852	PROJECT-2	Core Deptt.	0—0--12	350		250	600	12
	TOTAL				560	60	380	**1000**	**24**

Specialization : ADVANCE CARPET TECHNOLOGY(ACT),
HOME TEXTILE TECHNOLOGY(HTT),
TEXTILE DESIGN TECHNOLOGY(TDT)

TEXTURED YARN TECHNOLOGY (NPTEL/MOOCS) 3-1-0 CREDIT 4

- Introduction, general classification of textured yarns
- False twist texturing, principles mechanism
- False twist texturing process parameters
- Draw-texturing, need
- Draw –texturing process parameters
- Friction texturing principles
- Friction texturing NCV drives
- Air Jet Texturing
- Air jet texturing process parameters
- Interlacement; need and principles, Bulked continuous filalament (BCF) yarns
- Hi-bulk yarns, principle and processes
- Solvent and chemical texturing

ADVANCE FABRIC STRUCTURE 3-1-0 CREDIT 3

UNIT –I **[8]**
Extra warp and extra weft designed fabric.
UNIT –II **[8]**
Damask and brocade fabric.

UNIT –III **[8]**
Warp backed and weft backed fabric. Swivel and lappet fabric design.

UNIT IV **[8]**
Double cloths
Introduction, classification, stitched double cloths, interchanging double cloths.
Treble cloth

Unit V
Tapestry structures- Introduction, simple weft face tapestries, combined warp & weft tapestries.

References:
1. Watson's Textile Design and Colour by Z Grosicki; Universal Publishing Corporation, Bombay (India)
2. Grammer of Textile Design – Nisbet
3. Structural Fabric Design by – Kilby
4. Woven Structures and Design – Doris Goerner; British Textile Technology GroupWIRA House, Leeds (UK)
5. Fibre to Fabric by Ghosh
6. Watson's Advance Textile Design

ADVANCE TEXTILE PRINTING TECHNOLOGY (NPTEL/MOOCS) 3-0-0 CREDIT 3

- Recall conventional printing, colourants: dyes and pigments
- Thickeners, viscosity, rheology, discharge –resist style
- Transfer printing, transfer paper printing, dyes and inks
- Mechanism of sublimation transfer, free path length, general machinery
- Introduction to digital printing, ink-jet printing principles: continuous jetting
- Drop-on-demand principle, general machinery
- Pre-treatment, ink-types and substrate, pigment inks
- Disperse inks, acid inks, reactive inks

QUALITY CONTROL IN HOME TEXTILES 3-0-0 CREDIT 3

UNIT – I [8]

Introduction to Quality .Importance of Quality Managing Quality through inspection, Managing Quality through testing. Seven tools Of Quality Control. Current Concepts in Quality Management.

Raw Material Inspection: Fabric Inspection, 4-point system, fabric defects, sewing threads, zippers,

UNIT – II [8]

Inspections Techniques: - In process Inspection**:** Introduction, spreading, cutting, sewing, pressing/finishing, fusing, screen printing, embroidery.Sewing Problems Problem of Stitch Formation, Problem of Pucker, Problems of Damage to the Fabric along the stitch line Final Inspection. How much to inspect., Selection of Inspectors.

UNIT – III [8]

Textile Testing & Production Evaluation.

Precision & Accuracy of Test methods Atmospheric Conditions for Testing, Strength Properties of apparel, fabric stretch properties , Dimensional Changes in Soft Furnishing due to laundering , dry cleaning ,steaming and pressing , Durable press evaluation of Fabrics & Soft Furnishings. Needle cutting & Yarn severance, Sew ability of Fabrics, Bow & Skew ness, Fabric Distortion, Fabric streak, Soil / Stain Releasing Tests.

UNIT – IV - Textile Testing & Production Evaluation [8]

Bonded & laminated furnishing Fabric, Testing of Fusible interlining. Testing of Zippers, Elastics Waist band Testing Yarn Strength & Elongation, Sewing Threads, Buttons & Snap Fastness & Wear Testing, Flammability of Clothing, Textiles Flammability Testing Methods, Factors Affecting Fabric flammability.

UNIT-V: Shade Sorting & Care Labeling [8]

Introduction to shade Sorting. Fundamental Color and color measuring, Instrumental Shade sorting.International care labeling System.

Grand total lectures required=40

Reference:

1. Managing quality control in apparel industry by P.V.Mehta&S.K.Bharadwaj, ASQC Quality Press, Newyork, 1998.
2. An Introduction to quality control for the apparel industry by P.V.Mehta, ASQC Quality Press, New York.
3. Principles of textile testing by J.E.Booth,C.B.S., publishers and distributers,New Delhi,1996
4. Process control in home textile manufacturing by K Kgoswami, Abhishek Publisher Chandigarh, India.(In press)

Design Management 3—0---0 CREDIT 3

UNIT I [8]

Difference between design and development, Aspects of home textiles design, Sources of design ideas / inputs, Design requirements pertaining to different countries, Characteristics of industry during various phases of product life cycle

UNIT II [8]

Stages in Textile Design process (Design planning): – Design output, Design inputs, Selection of raw material and accessories, Customer involvement in design process, Innovation in the design process

UNIT III [8]

Design for manufacturing (DFM), Quality function deployment (QFD), Design and development system requirements as per ISO, designing for future

UNIT IV [8]

Validation and verification of designs, Time management in designing, Basics of network analysis (PERT & CPM), Standard costing methods professional practices of designing,

UNIT V [8]

Presentation of designsneed and maintenance, Design related records and their maintenance, Design catalogue – preparation, Evaluation of performance of designs; Define Elements and their Aesthetic value and marketing prospect.

References:

1. Design Management by Brigitte de Mozota
2. Design Project Management by Griff Boyle
3. Principles and Practice of Management by Prasad, L M.
4. Principle of Management by Tripathi & Reddy, P.N.
5. Production Operation Management by Heizer, Joy.
6. Export, Import Procedure & Documentation by Jain, Archarya.
7. Textile Designing: Theory & Concept by Jain, Tanya

8. Watson's Textile Design and Colour by Grosicki, Z. J.
9. Hand book of textile design, Jacquie Wilson, wood head, publishing UK

RCT851	**GD AND SEMINAR (0—0—3)**	**CREDIT:2**
RCT852	**PROJECT 2 (0—0—12)**	**CREDIT:12**